3-D Gabor Convolutional Neural Network for Hyperspectral Image Classification

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***Abstract*— Due to the detailed spectral information through hundreds of narrow spectral bands provided by hyperspectral image (HSI) data, it can be employed to accurately classify diverse materials of interest, which is one of the core applications of hyperspectral remote sensing technology. In recent years, with the rapid development of deep learning, convolutional neural networks (CNNs) have been successfully applied in many fields, including HSI classification. However, the random gradient descent-based parameter updating scheme is too general and leading to the inefficiency of CNN models. Moreover, the high dimensionality and limited training samples of HSI data also exacerbate the overfitting problem. To tackle these issues, in this article, a novel deep network with multilayer and multibranch architecture, named 3-D Gabor CNN (3DG-CNN), is proposed for HSI classification. More precisely, since the predefined 3-D Gabor filters in multiple scales and orientations could well characterize the internal spatial–spectral structure of HSI data from various perspectives, the 3-D Gabor-modulated kernels (3-D GMKs) are employed to replace the random initialization kernels. Moreover, the specially designed multibranch architecture enables the network to better integrating the scalable property of 3-D Gabor filters; thus, the representative ability and robustness of the extracted features can be greatly improved. Alternatively, the number of network parameters is substantially reduced due to the incorporation of 3-D Gabor modulation, relieving the training complexity and also alleviating the training process from overfitting. Experimental results on four real HSI datasets (including two newly released ones in the literature) have demonstrated that the proposed 3DG-CNN model can achieve better performance than several widely used machine-learning-based and deep-learning-based approaches. For the sake of reproducibility, the codes of the proposed 3DG-CNN model are available at http://jiasen.tech/papers/.**

***Index Terms*— Convolutional neural networks (CNNs), Gabor wavelet, hyperspectral image (HSI).**

NOMENCLATURE

Ψ	3-D Gabor filters.
f	Scale factor of 3-D Gabor filters.
φ and θ	Orientation factors of 3-D Gabor filters.
U and V	Number of scales and orientations.
F and $\hat{F}$	Input and output feature maps.
I and O	Number of input and output feature maps.
C	Standard CNN kernel (SCK).
G	3-D Gabor modulated kernel (3-D GMK).
N and W	Kernel size of SCK in spectral and spatial domains.
B	Number of spectral bands of HSI data.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) captured by hyperspectral sensors often contain hundreds of narrow spectral bands ranging from visible to near-infrared. The abundant spectral information provided in these bands is highly correlated with the intrinsic properties of different materials. With this rich information, it is possible to accurately characterize and identify the materials of objects of our interest. Due to this advantage, HSIs have been utilized in a variety of applications, such as agriculture, mineralogy, surveillance, physics, astronomy, chemical imaging, and environmental sciences [1]. However, detailed spectral information provided by hyperspectral sensors presents both opportunities and challenges, in which the complexity of spectral and spatial structures, high dimensionality, and limited training samples may lead to the Hughes phenomenon [2]. Therefore, the classification of HSI data remains a challenging task.

There are currently multitudinous classification methods for HSI data processing. Most of them focus on discriminative spatial, spectral, or joint spatial–spectral feature extraction. For example, the principal component analysis (PCA) has been widely used for the hyperspectral analysis [3], which

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captures the maximum variance of the signal by projecting the signal onto the eigenvectors of the covariance matrix. In [4], the linear discriminant analysis (LDA) was exploited to reduce the dimension of HSI data before classification, after which discriminant analysis feature extraction became more and more popular in HSI classification. Further attempts using graph-embedding-based discriminant analysis (GDA) and its extensions are present in [5]. Besides, low-rank representation and sparse representation have been utilized to characterize HSI feature in [6] and [7]. In [8], a pseudolabel guided kernel learning approach has also been performed to classify HSI data. Apart from these, among the existing feature extraction techniques for HSI data, 2-D Gabor filtering has attracted a lot of attention due to its capability to extract discriminative and informative edge and texture information with extractors of different scales and orientations [9]–[11]. However, these approaches only utilize the information in the spatial domain, while the correlation between bands is unconsciously ignored. To fully utilize both the spatial information and spectral correlation of HSI data, 3-D Gabor filtering, which is more suitable for tensor-structure data, was introduced as an effective way to obtain the spectral–spatial features from HSI data and has shown advantageous performance [12]–[15].

In recent years, following the great success in traditional optical RGB image classification tasks, deep-learning-based (DL-based) methods have been applied for HSI classification problems and achieved state-of-the-art performance. In [16], the stacked autoencoder (SAE) deep model was first proposed for HSI classification. After that, many DL-based approaches have been developed, especially convolutional neural networks (CNNs). In [17], a 1-D CNN model was employed to extract spectral features of each pixel, while, in [18], a 2-D CNN was explored to extract spatial features from a local cube with 2-D convolutions in the spatial domain. Yet, neither of these two approaches makes the best use of the rich spatial–spectral correlation of HSI data. Thus, in [19], learning sensor-specific spatial–spectral features of HSI data were extracted by CNN. Furthermore, in [20], a promising approach directly handles the hyperspectral cube with 3-D CNNs, which works simultaneously on both spatial and spectral dimensions using 3-D convolutions.

Different from traditional machine-learning-based (ML-based) methods, which require careful engineering and expert domain knowledge, CNNs are designed to automatically learn high-level features from raw data in a hierarchical manner. These features are more discriminative, abstract, and robust than those in traditional methods. However, CNN models often require a large number of training samples in order to avoid overfitting due to the large number of free parameters involved, which is more severe for HSI data with high dimensionality and lack of labeled data. Moreover, the parameter learning procedure of a CNN is typically nonconvex, which is usually addressed with a gradient descent algorithm under a random scheme, potentially missing some important information, such as representative features at different scales and orientations.

To tackle these issues and further promote the performance of CNNs, *a priori* knowledge is embedded into original CNN architectures. More specifically, Gabor filtering with enhanced capacity to extract discriminative spatial–spectral features of different scales and orientations is already utilized as *a priori* knowledge in a few preprocessing models to reinforce CNNs. For example, in [21]–[23], Gabor filters are first applied to extract internal features of original image data, after which the extracted features are used as the input of the CNN module. Further attempts have also been made to employ Gabor filters in the convolutional layers of CNNs. In [24]–[26], researchers replace the regular random kernels in the first or second convolutional layer with fixed Gabor filters to reduce the number of parameters. Therefore, only the remaining layers of the CNN models are trained. These methods, to a certain degree, relieve the training complexity of CNNs, yet they do not explicitly integrate the Gabor filter into the convolution filters. In fact, many filters in CNNs, especially those in the first several layers, look very similar to the Gabor filters [27]. In [27], a strategy different from the above techniques is introduced by modulating regular convolutional kernels with the predefined Gabor filters and has achieved great reinforcement of traditional CNN models. In particular, it is worth pointing out that all of these methods involve only 2-D Gabor filters. However, for HSI data that is of a tensor nature, involving 2-D Gabor wavelets into CNNs is insufficient to fully characterize the rich spatial–spectral information as mentioned before.

Inspired by the aforementioned observations, we purpose to jointly utilize the enhanced scale and orientation analysis and interpretation ability of 3-D Gabor filtering and the formidable learning capability of CNNs, so as to guide the CNN modules to generate features with an enhanced robustness against the orientation and scale changes and reduce the training complexity of traditional CNNs in the meantime. In this article, a novel deep learning network integrating 3-D Gabor filtering and convolutional kernels of traditional CNNs, named 3-D Gabor CNN (3DG-CNN), is proposed for pixelwise classification in HSIs. First, multiscale and multiorientation 3-D Gabor filters are predefined; 13 different orientations of 3-D Gabor filters are involved to cover the whole 3-D space of Gabor orientations. Second, the steerable properties of these 3-D Gabor filters are inherited into the CNNs by modulation, which carries out element-by-element multiplication between 3-D Gabor filters and standard CNN kernels (SCKs), generating 3-D GMKs. Third, the GMKs are used to replace the SCKs of each convolutional layer in our network. Then, we implement the update of the 3-D GMKs in the backpropagation process by a modified formulation. Finally, we design a multilayer and multibranch network to make the best of the multiscale feature extraction capacity of 3-D GMKs.

For each convolutional layer, multiscale and multiorientation 3-D Gabor filters are predefined and modulated with initial CNN kernels to generate 3-D GMKs, which are used to replace the initial CNN kernels. On the one hand, the special multibranch design enables the network to better integrate the scalable property of 3-D Gabor filters into SCKs, and the network is capable to learn features with more robustness against orientation changes through involving all 13 orientations of Gabor filters. Besides, compared with the 2-D Gabor wavelet, with the involvement of the 3-D Gabor wavelet, the network

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is more talented to extract spatial and spectral features from the HSI data cube, which is of great effectiveness in HSI classification. On the other hand, since the 3-D Gabor filters are already predefined, only the SCKs are learned during the training procedure, which greatly reduces the network parameter to be trained, relieves the training complexity, and prevents the training process from overfitting to a certain extent.

Before presenting the detailed contributions of the proposed 3DG-CNN approach, the technical differences from [27] have been clarified. First, with respect to the spatial–spectral structure of hyperspectral data, we used multiple 3-D Gabor kernels to cooperate with the CNN model. Compared with the modulated kernels using original 2-D Gabor kernels in [27], 3-D GMKs in our 3DG-CNN approach can better characterize the joint spatial–spectral information inside hyperspectral data. Moreover, the modulation procedure of 3-D GMK is different from that of 2-D Gabor kernels in [27]. Second, with respect to the high-dimensional structure of hyperspectral data, the input of 3DG-CNN is normalized as independent patches of 13 bands to adapt with the dimensionality of the filter size. In [27], multiple copies of all bands are used as input, and more parameters should be tuned. Alternatively, our 3DG-CNN divides the whole HSI cube into separated subsets. Furthermore, in order to further improve the performance of our network, we reduced the number of network layers and adopted a multibranch network structure different from [27], which can make more effective use of the multiscale characteristics of the multiple 3-D Gabor kernels. More specifically, the main contributions of this article can be summarized as follows.

1) It is the first attempt in HSI classification to combine multiscale and multiorientation 3-D Gabor operators with SCKs by modulation in each convolutional layer. 3-D Gabor wavelet, compared to 2-D Gabor wavelet, is much more suitable for HSI data to extract spatial–spectral features, which usually results in a more excellent performance. The modulated CNN kernels lead the CNNs to extract more discriminative features due to the ameliorated scale and orientation analysis and interpretation ability of 3-D Gabor filtering, which may probably be insufficient in SCKs.
2) With all the 13 orientations of the 3-D Gabor filter involved in the modulation process that could cover the entire 3-D tensor space, the resulting 3-D GMKs are more talented to extract multiorientation features from the HSI cube. Meanwhile, the special multibranch network structure design, which corresponds to the multiscale 3-D Gabor convolution, enables the extract features more robust to scale changes.
3) The proposed network is light and easy to be trained. During the learning procedure of the network, since the 3-D Gabor filters have already been predefined, only the parameters of standard convolutional kernels (SCKs) should be updated and saved. This reduces the number of trainable parameters and relieves the complexity of training, making the network light and compact, while preventing overfitting and preserving strong performance. Besides, through dividing the whole HSI cube into separated subsets, 3DG-CNN reduces not only the number of parameters but also the sliding operations of SCKs in the spectral dimension, and thus, the complexity of the training procedure can be greatly alleviated. For the sake of reproducibility, the codes of the proposed 3DG-CNN model are available at http://jiasen.tech/papers/.

In order to show the effectiveness of the proposed 3DG-CNN approach, a number of experiments have been carried out on four real HSI datasets, including two classic ones (i.e., the Houston 2018 and Pavia University) and two newly released ones (i.e., the Matiwan Village and Yellow River Estuary). Experimental results consistently illustrate that the proposed 3DG-CNN model can achieve better performance than several widely used ML- and DL-based approaches. The descriptions of all the mathematical notations used in this article are summarized in Nomenclature.

The remainder of this article is organized as follows. Section II makes a brief introduction to the 3-D Gabor filter and CNNs. In Section III, the proposed 3DG-CNN method is described with four parts in detail. Afterward, Section IV elaborates the experiment settings and results systematically. Finally, in Section V, we conclude this article with some remarks.

II. Related Works

A. 3-D Gabor Filter

Since the frequency and orientation descriptions of conventional 2-D Gabor filter are similar to those of human visual system [28], [29], the 2-D Gabor filter is particularly suitable for texture representation and discrimination in the spatial domain and has been widely applied to extract features for face recognition [30], [31], texture classification [32], gait recognition [33], and so on. Furthermore, in order to retain the inherent physical properties of the HSI data and reveal the signal changes in joint spatial–spectral domains, the Gabor filter has been extended to three dimensions, which can be directly used to extract features from volume data and has shown good classification performance [12]–[15]. In particular, a 3-D Gabor filter is a sinusoidal wave modulated by a 3-D Gaussian function. Specifically, for a certain pixel at the spatial–spectral position (α, β, γ), a circular 3-D Gabor filter can be defined as follows:

$$\Psi_{f,\varphi,\theta}(\alpha,\beta,\gamma) = \frac{1}{(2\pi)^{\frac{3}{2}}\sigma^3} \times \exp(j2\pi(\alpha x + \beta y + \gamma z)) \times \exp\left(-\frac{\alpha^2+\beta^2+\gamma^2}{2\sigma^2}\right) \quad (1)$$

where $x = f\sin\varphi\cos\theta$, $y = f\sin\varphi\sin\theta$, and $z = f\cos\varphi$. Variable f is the central frequency of the sinusoidal wave, φ and θ represent the angles of the wave vector f with the z-axis and the xy plane in the frequency domain (x, y, z) (see Fig. 1), and σ is the width of Gaussian envelop in the (α, β, γ) domain.

B. Convolutional Neural Networks

Over the past decade, CNNs have shown remarkable performance in image classification [34], object detection [35],

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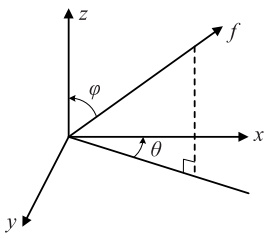

Fig. 1. Illustration of the 3-D Gabor filter in the 3-D frequency domain.

semantic segmentation [36], and so on. Due to the large flexibility of CNNs and their ability to make strong assumptions about the input images, CNNs have also been applied to the HSI data and become one of the most successful tools for HSI data classification [37]. Typically, the computation process of a CNN can be divided into two parts: forward computation and backward computation.

1) Forward Computation Process: A CNN model for classification typically consists of convolutional layers, pooling layers, and fully connected (FC) layers. In the convolutional layers, each convolution neuron processes data only for its receptive field to obtain the feature map for the next layer, which is similar to the response mechanism of neurons in the visual cortex to visual images. Therefore, convolutional layers have great advantages for extracting deep abstract features of image data. The forward computation process of convolutional layers is to convolve the input feature map with convolutional kernel channel by channel and perform an accumulation operation to obtain the output feature map. Formally, the input feature maps F and the output feature maps $\hat{F}$ of a convolutional layer can be denoted as

$$F = \{F_i | i = 1, 2, \ldots, I\} \tag{2}$$

$$\hat{F} = \left\{\hat{F}_o | o = 1, 2, \ldots, O\right\} \tag{3}$$

where I and O are the numbers of the input and output feature maps; F_i and $\hat{F}_o$ represent the ith input feature map and the oth output feature map, respectively. Then, the oth output feature map $\hat{F}_o$ can be obtained by the following convolution:

$$\hat{F}_o = \sum_{i=1}^{I} F_i \otimes C_{i,o} \tag{4}$$

where $C_{i,o}$ refers to the channel of convolutional kernel corresponding to $\hat{F}_o$ and F_i.

2) Backward Computation Process: CNNs, including all other DL-based models, learn model parameters by minimizing the loss function with the gradient descent method and the backpropagation (BP) algorithm. After the forward computation of all the layers, the gradients of the loss function for each parameter are computed with BP algorithm, and parameters are then updated in the opposite direction of the gradients, which can be formulated as follows:

$$\delta = \frac{\partial L}{\partial C_{i,o}} \tag{5}$$

$$C_{i,o} = C_{i,o} - \eta\delta \tag{6}$$

where L is the loss function, δ is the gradient of the loss function computed by chain rule, and η is the learning rate.

III. 3-D Gabor Convolutional Neural Network

Concerning hyperspectral data, due to the spatial regularity and the spectral similarity of surface materials, it is more desirable to extract joint spectral–spatial features for classification. Different from 2-D Gabor filters that extract only spatial signal correlations, 3-D Gabor filters can effectively characterize signal variation in joint spatial–spectral domains. In this article, in order to make full use of the advantages of both the CNN model and Gabor filtering, we introduce the 3-D GMKs, which is the first attempt in the literature to integrate together with the 3-D Gabor filters and the SCKs into CNN models. Then, a specially designed multibranch 3DG-CNN using 3-D GMKs was proposed for the HSI classification tasks.

In what follows, first, we give the details on how we obtain the 3-D GMKs through 3-D Gabor filters. Second, we describe the forward and backward propagation process of deep CNN models using 3-D GMKs with enhanced scale and orientation robustness. Third, we depict the architecture of our 3DG-CNN designed for HSI classification tasks.

A. 3-D Gabor-Modulated Kernels

As mentioned before, 3-D Gabor filters are steerable and manipulable with adjustable orientations and scales. In order to incorporate the steerable attributes into the Gabor CNN, a set of 3-D Gabor filters with different orientations and scales are predefined, denoted as

$$\Psi = \left\{\Psi_{u,v} | u = 1, 2, \ldots, U, v = 1, 2, \ldots, V\right\} \tag{7}$$

where U and V are the number of scales and orientations, respectively. To cover the entire 3-D space as much as possible and obtain the features in all directions, 3-D Gabor filters with different orientations can be designed by changing the value of parameter θ and φ. In this article, with $\pi/4$ as the step size, parameters θ and φ are, respectively, selected as

$$\{\theta, \varphi\} \in \left\{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}. \tag{8}$$

It is easy to notice that the frequency vector points to the same direction with different θ's when $\varphi = 0$; hence, there are in total 13 different orientations, i.e., $V = 13$. Moreover, to enhance multiscale feature representation, four different scales are selected in this article, i.e., scale factors f are selected as

$$f \in \{0.5, 0.25, 0.125, 0.0625\} \tag{9}$$

which varies from coarse to fine and could well characterize the texture features of surface materials (i.e., $U = 4$).

For the modulation procedure, 3-D GMKs are created by manipulating 3-D Gabor kernels on the SCKs to produce the scale and orientation enhanced feature maps. Before being modulated by 3-D Gabor kernels, SCKs are first learned by the BP algorithm. Unlike standard kernels in CNNs, the SCKs in 3DG-CNN are 3-D to encode the orientation channel of 3-D Gabor filters. Suppose that a standard kernel in CNNs is of size $W \times W$, and the size of an SCK in 3DG-CNN can be expressed as $N \times W \times W$, where $W \times W$ refers to the size of the kernel and N represents the channel. Furthermore,

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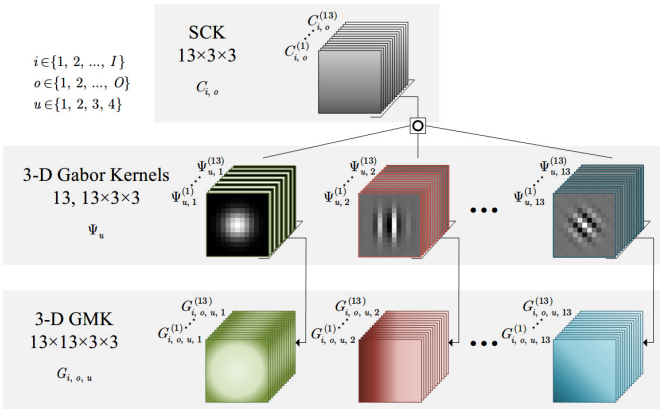

Fig. 2. Modulation process of a 3-D GMK using a 3-D SCK and a set of the uth scale 3-D Gabor kernels. The learned SCK is multiplied with 3-D Gabor kernels correspondingly in different orientations to produce the modulated subkernels. All $V = 13$ subkernels together form an integral 3-D GMK.

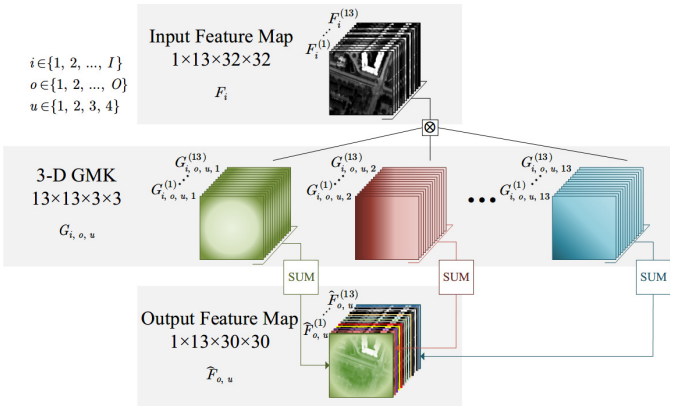

Fig. 3. Illustration of 3-D GMK convolution of a 13-channel input feature map with a 3-D GMK of the uth scale. The input feature map is convolved with the 3-D GMKs in each direction according to the channel, and all channels in each direction are summed together to obtain each channel of the output feature map, i.e., the vth direction of 3-D GMK corresponds to the vth channel of the output feature map.

the dimensions of the weight per layer in 3DG-CNN can be represented as $O \times I \times N \times W \times W$, where O and I refer to the channel of output and input feature map, respectively. For implementation convenience and to keep the consistency and continuity of the channel quantity of the feature maps, N is set to be the same as V, which is the number of orientations of the 3-D Gabor filters. A GMK is obtained based on a modulation procedure by manipulating an SCK via a 3-D Gabor filter bank consisted of V 3-D Gabor filters for the uth scale. Equation (10) and Fig. 2 show the details of the kernel modulation process. With respect to the vth orientation and the uth scale 3-D Gabor filter $\Psi_{u,v}$, the modulated subkernel $G_{i,o,u,v}$ of an SCK $C_{i,o}$ can be calculated by

$$G^{(n)}_{i,o,u,v} = C^{(n)}_{i,o} \circ \Psi^{(n)}_{u,v} \tag{10}$$

where i and o refer to the ith input feature map and the oth output feature map, (n) represents the nth channel, and $\circ$ is an element-by-element multiplication operation. Then, a 3-D GMK for a given scale u can be defined as

$$G_{i,o,u} = \left\{ G_{i,o,u,v} | v = 1, 2, \ldots, V \right\}. \tag{11}$$

It can be seen from Fig. 2 that a 3-D GMK consists of V 3-D subkernels ($V = 13$), and the size of a 3-D GMK is $V \times N \times W \times W$. Yet, under actual circumstances, only $N \times W \times W$ parameters of SCKs will be saved, as the 3-D Gabor filters are already predefined. That is to say, we are able to get enhanced feature maps by using GMKs without increasing the number of parameters. The capability of multiple 3-D Gabor kernels is integrated by learning a single 3-D SCK, which reduces the training parameters while maintaining superior classification ability. It is noteworthy that, since the CNNs are usually fulfilled with real-valued convolutions, we only use the real parts of 3-D Gabor filters in the modulation process defined by (1).

B. Forward Computation Process

In the forward computation procedure of deep models using 3-D GMKs, 3-D GMK convolution is performed on the input

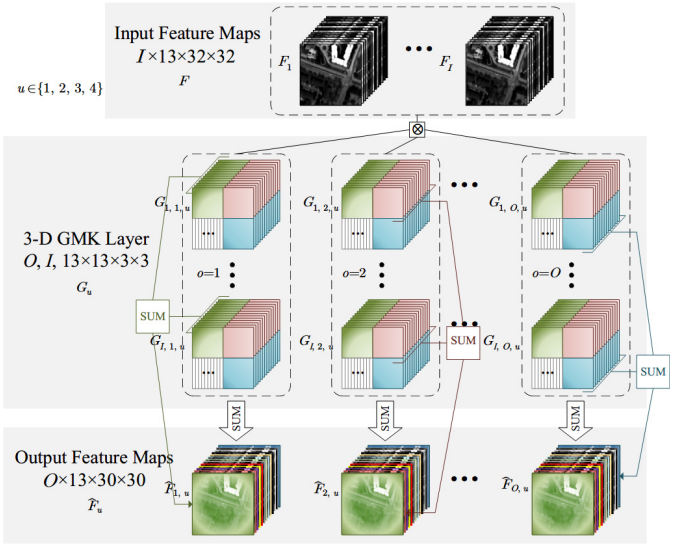

Fig. 4. Forward convolution process of a 3-D GMK convolutional layer with I input feature maps and O output feature maps, each of which is of $N = 13$ channels. There are O groups of 3-D GMKs, corresponding to O output feature maps. First, each channel of the input feature maps is convolved with the 3-D GMKs of the corresponding channel in each group, which generates I temporary feature maps for each group. Then, the temporary feature maps of each group are summed following the corresponding orientations to generate the output feature maps.

feature maps to produce output feature maps with scale and orientation robustness enhanced. Similar to traditional CNN, the input and output feature maps of a 3-D GMK convolutional layer can be expressed as (2) and (3). Furthermore, we denote the ith input feature map and the oth output feature map of a 3-D GMK convolutional layer as

$$F_i = \left\{ F^{(n)}_i | n = 1, 2, \ldots, N \right\} \tag{12}$$

$$\hat{F}_{o,u} = \left\{ \hat{F}^{(n)}_{o,u} | n = 1, 2, \ldots, N \right\} \tag{13}$$

which means that each input and output feature map of 3DG-CNN are of N channels, corresponding to V orientations of 3-D Gabor filters, respectively (i.e., $N = V = 13$).

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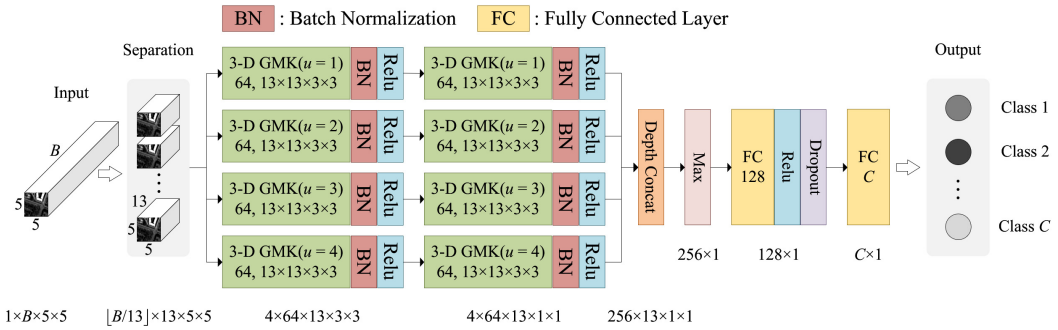

Fig. 5. Detailed architecture of the proposed 3DG-CNN.

u demonstrates that the output feature map is generated by the 3-D GMK at the uth scale. Then, for all I input feature maps F that each has N channels, the convolution procedure to obtain each channel of $\hat{F}_o$ can be formulated as

$$\hat{F}_{o,u}^{(v)} = \sum_{i=1}^{I} \sum_{n=1}^{N} F_i^{(n)} \otimes G_{i,o,u,v}^{(n)} \tag{14}$$

where (v) represents the vth channel of every output feature map $\hat{F}_{o,u}$, and (n) represents the nth channel of every input feature map F_i and subkernel $G_{i,o,u,v}$. Intuitively, Fig. 3 illustrates an example of the convolution process of a 3-D GMK on a single input feature map with $N = 13$ channels, whereas Fig. 4 presents the forward convolution process of a 3-D GMK convolutional layer with multiple input and output feature maps.

C. Backward Computation Process

As mentioned above, in 3DG-CNN, only the weights of SCKs need to be learned by the BP algorithm and saved after the forward calculation process since the 3-D Gabor filters are already given. We use chain rule to compute the gradients of the loss function for each parameter in each SCK, i.e., to sum up the gradients of all the subkernels in each 3-D GMK, which can be formulated as

$$\delta = \frac{\partial L}{\partial C_{i,o}} = \sum_{v=1}^{V} \frac{\partial L}{\partial G_{i,o,u,v}} \circ \Psi_{u,v} \tag{15}$$

where L is the loss function and δ refers to the gradient. After that, the weights of SCKs are updated on the contrary direction of the gradients according to (6). During the backward computation, only the partial derivative of SCKs is calculated, which is easy to implement and is the same as standard CNNs.

D. Network Architecture

The detailed architecture of the proposed 3DG-CNN model is illustrated in Fig. 5. Empirically, the patch size of the input feature map was set to be 5×5. The kernel size of 3-D GMK is $13 \times 3 \times 3$, and the step size of 3-D GMK is set to be 1. The layer width of each 3-D GMK layer was set to be 64, and no padding was adopted in the convolution. After each layer of the 3-D GMK + BN + ReLU structure, the spatial size of the output feature map is reduced by 2. Therefore, a two-layer structure was used to make the size of the output feature map of convolutional layers be 1×1. To adapt to the dimensionality of the filter size and maintain the coupling of the input feature maps and the 3-D GMKs, the input of 3DG-CNN was normalized as independent patches of 13 bands. Through dividing the whole HSI cube into separated subsets, 3DG-CNN reduces not only the number of parameters to be learned but also the sliding operations of SCKs in the spectral dimension ($\lfloor B/13 \rfloor$ versus $B-13$, where B is the number of spectral bands of HSI data), and thus, the complexity of training procedure can be further alleviated. Besides, a four-branch architecture of 3-D GMK convolutional layers was adopted, which is corresponding to the four scales of 3-D Gabor filters. With this multibranch architecture design, it is able to make use of the scalable property of 3-D Gabor filters in the learning procedure and extract features with multiscale robustness enhancement. In each 3-D GMK, all $V = 13$ orientations of 3-D Gabor filters are involved; hence, the extracted features are also enhanced for multiorientation representation. We used cross entropy as the loss function, which is widely used in deep learning networks to deal with the multiclassification problem. Formally, it can be defined as

$$L = -\sum_{x \in \mathcal{X}} \mathbb{P}_{\text{label}}(x) \log \mathbb{P}_{\text{pred}}(x) \tag{16}$$

where $x \in \mathcal{X}$ refers to the training samples. $\mathbb{P}_{\text{label}}$ and $\mathbb{P}_{\text{pred}}$, respectively, indicate real sample distribution and predicted probability distribution computed by a softmax layer. Furthermore, we adopted batch normalization (BN) and ReLU activation function after each 3-D GMK convolutional layer and the first FC layer. A dropout layer [38] was also adopted after the first FC layer to avoid overfitting.

IV. Experiments

A. Datasets

To evaluate the effectiveness of the proposed 3DG-CNN method for HSI classification, we used four real-world HSI datasets in the experiments, including three high-resolution datasets (i.e., the Pavia University, Houston 2018, and Matiwan Village) and a middle-resolution dataset (i.e., the Yellow River Estuary). It is worth mentioning that both the Matiwan Village and Yellow River Estuary are two recently released HSI datasets, which are very challenging in the literature.

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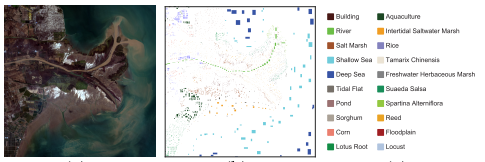


(a) (b) (c)

Fig. 6. (a) False-color map, (b) ground-truth map, and (c) labels of the Yellow River Estuary dataset.

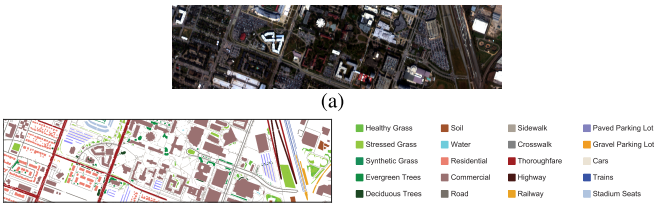
TABLE I

LAND COVER CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR THE YELLOW RIVER ESTUARY DATASET

Class	Land Cover Type	Training	Test
C1	Building	50	483
C2	River	50	5,326
C3	Salt Marsh	50	4,945
C4	Shallow Sea	50	17,500
C5	Deep Sea	50	18,627
C6	Intertidal Saltwater Marsh	50	2,293
C7	Tidal Flat	50	1,742
C8	Pond	50	1,737
C9	Sorghum	50	596
C10	Corn	50	1,459
C11	Lotus Root	50	2,669
C12	Aquaculture	50	7,969
C13	Rice	50	5,458
C14	Tamarix Chinensis	50	1,170
C15	Freshwater Herbaceous Marsh	50	1,367
C16	Suaeda Salsa	50	824
C17	Spartina Alterniflora	50	530
C18	Reed	50	1,920
C19	Floodplain	50	297
C20	Locust	50	25
	Total	1,000	76,937

1) Yellow River Estuary Dataset: The first hyperspectral dataset was acquired by the visible shortwave infrared advanced hyperspectral imager (AHSI), which is designed by the Shanghai Institute of Technical Physics, Chinese Academy of Sciences (CAS), and mounted as the main payload on the platform of China's Gaofen5 satellite [39]. The image scene was captured on November 1, 2018, over the Yellow River estuary area located in the northeast of Shandong province, China, mainly distributing wetland vegetation. The AHSI sensor collects as many as 330 spectral bands ranging from 400 to 2500 nm, including 150 VNIR bands (400–1000 nm) and 180 SWIR bands (1000–2500 nm). After discarding some bad bands, the remaining 280 bands are used in the experiments. This dataset covers a large scene of 1400 × 1400 pixels with a spatial resolution of 30 m per pixel. There are 77 937 labeled samples from 20 different land cover types in the image (see Fig. 6), of which the training and test portions are exhaustively provided in Table I.

2) Houston 2018 Dataset: This dataset was collected on February 16, 2017, with the compact airborne spectrographic imager from the ITRES company (ITRES-CASI 1500) over the area of the University of Houston campus, covering an urban area of over 4 km^2. It was first known and distributed by the Image Analysis and Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) and the University of Houston for the 2018 Data Fusion



(a) (b) (c)

Fig. 7. (a) False-color map, (b) ground-truth map, and (c) labels of the Houston 2018 dataset.

TABLE II

LAND COVER CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR THE HOUSTON 2018 DATASET

Class	Land Cover Type	Training	Test
C1	Healthy Grass	200	9,599
C2	Stressed Grass	200	32,302
C3	Synthetic Grass	200	484
C4	Evergreen Trees	200	13,388
C5	Deciduous Trees	200	4,848
C6	Soil	200	4,316
C7	Water	200	66
C8	Residential	200	39,562
C9	Commercial	200	223,484
C10	Road	200	45,610
C11	Sidewalk	200	33,802
C12	Crosswalk	200	1,316
C13	Thoroughfare	200	46,158
C14	Highway	200	9,649
C15	Railway	200	6,737
C16	Paved Parking Lot	200	11,275
C17	Gravel Parking Lot	100	49
C18	Cars	200	6,378
C19	Trains	200	5,165
C20	Stadium Seats	200	6,624
	Total	3,900	500,812

Contest [40], [41]. In this article, we utilized the training portion of the whole dataset. The spectral range of the dataset is 380–1050 nm with 48 bands and a spatial resolution of 1 m per pixel. The dataset contains 601 × 2,384 pixels and 20 land cover classes of interest (see Fig. 7). The training and test portions used in the experiments are given in Table II.

3) Matiwan Village Dataset: This aerial hyperspectral remote sensing dataset was collected using the visible and near-infrared imaging spectrometer designed by the Shanghai Institute of Technical Physics, CAS, in October 2017, over the Xiongan New Area, China [42]. The Matiwan Village dataset covers the spectral range of 400–1000 nm, with 250 bands and a geometric resolution of 0.5 m per pixel. The dataset contains 3750 × 1580 pixels and 20 land cover classes of interest, mainly cash crops. A total number of 3 677 110 samples from 20 land covers are labeled (see Fig. 8), of which the training and test sample numbers are clearly listed in Table III. It is clear that, due to the huge number of labeled samples used in the experiment, the Matiwan Village HSI is an excellent dataset to test the performance of the compared classification methods.

4) Pavia University Dataset: This airborne dataset was captured by the Reflective Optics Spectrometer (ROSIS) sensor over the University of Pavia, Northern Italy, in 2003. The Pavia University dataset contains 115 spectral bands, of which

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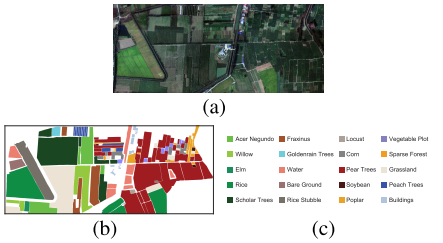
TABLE III

LAND COVER CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR THE MATIWAN VILLAGE DATASET

Class	Land Cover Type	Training	Test
C1	Acer Negundo	500	225,147
C2	Willow	500	180,266
C3	Elm	500	14,853
C4	Rice	500	451,644
C5	Scholar Trees	500	475,091
C6	Fraxinus	500	168,842
C7	Goldenrain Trees	500	22,804
C8	Water	500	165,147
C9	Bare Ground	500	37,909
C10	Rice Stubble	500	193,330
C11	Locust	500	5,112
C12	Corn	500	58,665
C13	Pear Trees	500	1,026,013
C14	Soybean	500	6,651
C15	Poplar	500	90,572
C16	Vegetable Plot	500	28,648
C17	Sparse Forest	500	996
C18	Grassland	500	421,290
C19	Peach Trees	500	65,014
C20	Buildings	500	29,116
	Total	10,000	3,667,110



(a) (b) (c)

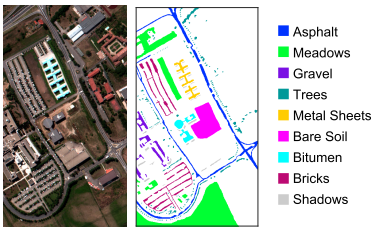
Fig. 8. (a) False-color map, (b) ground-truth map, and (c) labels of the Matiwan Village dataset.

12 noisy bands are usually omitted, and only 103 channels are kept for the experiments. The spatial size of the dataset is 610 × 340, and the ground-sampling distance is 1.3 m per pixel. The dataset has 42 776 labeled samples in total, and nine classes of land covers are considered (see Fig. 9). The number of training and test samples for different classes of interest used in the experiments is given in Table IV.

B. Experimental Setup

To purposefully illustrate the performance and verify the validity of our proposed 3-D-Gabor-CNN-integrated method (3DG-CNN), five ML-based classifiers and five advanced DL-based methods were utilized in comparison. All these methods are described in detail as follows.

1) *Support Vector Machine With Raw Hyperspectral Data (RAW-SVM):* The SVM classifier is applied on the raw HSI data without feature extraction.
2) *Support Vector Machine With Collaborative GDA (CGDA-SVM):* The SVM classifier is applied to the features of HSI extracted by CGDA.
3) *Support Vector Machine with LDA (LDA-SVM):* The SVM classifier is applied to the features of HSI extracted by LDA.



(a) (b) (c)

Fig. 9. (a) False-color map, (b) ground-truth map, and (c) labels of the Pavia University dataset.

TABLE IV

LAND COVER CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR THE PAVIA UNIVERSITY DATASET

Class	Land Cover Type	Training	Test
C1	Asphalt	150	6,481
C2	Meadows	150	18,499
C3	Gravel	150	1,949
C4	Trees	150	2,914
C5	Metal Sheets	150	1,195
C6	Bare Soil	150	4,879
C7	Bitumen	150	1,180
C8	Bricks	150	3,532
C9	Shadows	150	797
	Total	1,350	41,426

4) *Support Vector Machine With 2-D Gabor Features (2DG-SVM):* The SVM classifier is applied on the 2-D Gabor features of the original HSI cube, and PCA is performed on the extracted features to reduce the spectral dimension before input.
5) *Support Vector Machine With 3-D Gabor Features (3DG-SVM):* The 3-D Gabor features extracted from original HSI data and later processed by PCA are utilized as the input of the SVM classifier.
6) *2DCNN:* CNN using 2-D SCKs is designed for HSI classification with raw hyperspectral data as input [43]. It is constructed with three 2-D convolutional layers and an FC layer. Each convolutional layer is followed by a BN layer, an ReLU activation function, and an MP layer.
7) *3DCNN:* 3-D convolutional kernels are applied in the CNN model for the volumetric HSI data [44]. Four 3-D convolutional layers, two 1-D convolutional layers, and an FC layer are used in the model. The ReLU activation function is utilized after the first and third 3-D convolutional layers and both 1-D convolutional layers.
8) *Multibranch CNN (MB-CNN):* By merging multiple branches on an ordinary CNN, a multibranch fusion network, which introduces 1 × 1 convolutional layer and L_2 regularization, is used to extract features of HSI data [45].
9) *Multiscale CNN (MS-CNN):* A multiscale 3-D CNN is performed to classify HSI data by jointly learning both 2-D multiscale spatial feature and 1-D spectral feature of HSI [46].
10) *2-D Gabor CNN (2DG-CNN):* 2-D Gabor filtering is integrated with CNN for gray or RGB image classification [27]. To extend the model for HSI classification, raw hyperspectral data are first separated into patches and then fed as multichannel input for the network.

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TABLE V
CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE FOR THE YELLOW RIVER ESTUARY DATASET

Class	RAW-SVM	CGDA-SVM	LDA-SVM	2DG-SVM	3DG-SVM	2DCNN	3DCNN	MB-CNN	MS-CNN	2DG-CNN	3DG-CNN
C1	63.48	73.85	69.72	**89.39**	81.94	65.06	73.88	85.91	88.37	79.14	83.78
C2	97.70	99.75	99.25	96.58	98.98	99.67	98.00	99.37	98.69	**99.92**	99.47
C3	78.13	68.26	77.68	82.83	77.80	63.87	72.01	64.07	84.08	79.22	**85.46**
C4	89.94	93.01	87.09	91.42	93.95	89.31	91.29	91.00	87.20	94.41	**96.30**
C5	95.85	98.58	95.04	93.64	94.74	98.74	96.42	99.89	**100.00**	98.13	97.91
C6	70.00	79.93	75.43	82.17	72.94	85.38	81.76	70.37	86.66	**87.13**	85.35
C7	57.00	57.84	54.42	76.80	64.25	63.84	63.60	**95.92**	72.52	63.86	70.63
C8	87.43	89.21	88.98	74.66	**94.82**	87.32	89.29	89.97	88.99	88.90	90.72
C9	69.58	85.40	85.52	**92.66**	88.16	81.63	80.92	92.24	91.00	86.56	89.94
C10	74.93	86.27	79.04	80.72	81.53	76.96	78.29	85.46	85.42	83.32	**87.70**
C11	63.87	74.20	82.45	74.37	87.50	77.96	84.75	89.13	89.13	87.04	**89.51**
C12	69.56	77.41	83.67	71.83	83.50	81.43	85.60	89.50	90.00	86.41	**90.07**
C13	67.21	73.58	81.06	78.38	81.31	73.83	73.99	87.43	**88.39**	88.26	85.25
C14	89.13	90.27	72.94	74.45	78.81	89.22	87.27	94.52	**94.88**	92.08	92.56
C15	90.00	91.62	76.46	77.81	90.77	88.41	89.73	91.95	94.16	94.80	**95.04**
C16	81.27	94.30	87.14	89.37	82.82	87.29	90.66	**95.88**	93.61	93.36	94.23
C17	78.28	89.82	92.44	93.28	93.71	83.68	83.70	**95.37**	94.51	94.79	94.75
C18	64.87	62.69	50.86	52.80	59.62	55.70	71.75	75.18	81.07	69.96	**82.94**
C19	47.71	73.31	66.70	**91.97**	81.54	60.22	65.20	62.82	85.40	82.84	90.05
C20	95.99	99.20	95.20	84.03	**100.00**	99.74	99.48	**100.00**	**100.00**	99.48	97.31
OA	83.54	87.11	85.64	85.55	88.31	86.16	87.59	90.53	91.23	91.10	**92.75**
AA	76.60	82.93	80.05	82.46	84.44	80.46	82.88	87.80	89.70	87.48	**89.95**
κ	0.81	0.86	0.84	0.84	0.87	0.85	0.86	0.88	0.91	0.90	**0.92**

For the training setting of 3DG-CNN, we used the Adam optimizer [47] to update the parameters for 120 training epochs, with the learning rate and weight decay setting to 0.001. The learning rate was divided by 2 for every 20 training epoch. It should be pointed out that, except for the patch size of input setting to 5, the same as 3DG-CNN, the other settings of all compared DL-based methods used in experiments followed their original papers. For a fair comparison, the 2-D Gabor filters in 2DG-SVM and the 3-D Gabor filters in 3DG-SVM were the same as 2DG-CNN and our proposed 3DG-CNN, respectively, and the numbers of reduced dimensions by PCA of both methods were set to the number of land cover types of each dataset. Besides, the parameter setting of CGDA-SVM and LDA-SVM was the same as [43].

With respect to the performance quantification metrics, the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (κ) of the classification results were used in the experiments. Precisely, OA is computed by dividing the number of correctly classified samples by the total number of test samples, while AA is the average classification accuracy of all land cover categories. Furthermore, the Kappa coefficient is a statistic widely used to measure the agreement, especially in multiclassification tasks with unbalanced samples [48]. During all the experiments, an NVIDIA Tesla P100 GPU was employed.

Before presenting the detailed experimental results, it should be pointed out that our proposed 3DG-CNN is parameter saving. That is to say, with the predefined multiscale and multiorientation 3-D Gabor filters, we only need to learn a small set of filters (SCKs), which are then modulated to generate several times of filters (3-D GMKs). Taking the GMK of 13 directions, for example, we can decrease the width of the layer to $1/13$ to reduce the model size of 3DG-CNN, which could greatly reduce the burden of training and maintain sufficient learning ability. Meanwhile, the multibranch structure in 3DG-CNN (as displayed in Fig. 5) also guarantees the robustness of the proposed method.

C. Experiments With Fixed Number of Training Samples

In this part, the training and test samples were randomly selected from each dataset with fixed numbers according to Tables I–IV. In order to alleviate the influence of random factors, all the experiments were repeated ten times, and the mean values of OA, AA, and Kappa coefficient were reported in the experiments. The time required for training on the Yellow River Estuary, Houston 2018, Matiwan Village, and Pavia University datasets is about 40, 20, 100, and 5 min, respectively. Besides, the experimental results on the four datasets with ten compared methods, including five ML-based classifiers, five advanced DL-based methods, and the proposed 3DG-CNN approach, are reported quantitatively and visually.

First, we quantitatively evaluate the classification performance on the Yellow River Estuary dataset with the accuracy of each class and three forenamed metrics in Table V, where the proposed 3DG-CNN outperforms all the other approaches in most cases. Specifically, the accuracies obtained by RAW-SVM are lower than all the other methods, indicating the importance of feature extraction. The other four ML-based methods achieve similar results with 2DCNN and 3DCNN. With specially designed architectures or kernels, MB-CNN, MS-CNN, 2DG-CNN, and 3DG-CNN outperform all the other methods with a significant improvement, which can be attributed to the positive impact of the multibranch architecture, multiscale architecture, and the integration of Gabor filtering on CNN models. Furthermore, since 3-D

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TABLE VI
CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE FOR THE HOUSTON 2018 DATASET

Class	RAW-SVM	CGDA-SVM	LDA-SVM	2DG-SVM	3DG-SVM	2DCNN	3DCNN	MB-CNN	MS-CNN	2DG-CNN	3DG-CNN
C1	90.98	92.92	91.72	84.82	91.81	**97.26**	95.35	97.08	96.61	97.01	97.13
C2	87.14	84.95	81.17	66.24	83.92	85.60	88.20	88.79	**90.27**	88.99	88.98
C3	90.69	**100.00**	**100.00**	98.25	**100.00**	99.89	99.87	**100.00**	**100.00**	99.79	99.79
C4	90.48	92.41	94.47	62.98	94.40	93.69	94.67	94.98	**96.34**	95.86	95.74
C5	75.53	83.78	88.81	71.90	89.27	84.75	89.06	88.73	**91.61**	90.55	91.55
C6	69.15	96.53	97.74	86.82	**99.27**	97.09	97.78	98.67	98.48	98.25	98.27
C7	93.04	98.80	**100.00**	**100.00**	**100.00**	99.69	99.63	**100.00**	99.86	**100.00**	99.86
C8	59.97	71.77	71.21	67.22	**87.04**	74.79	70.75	77.90	77.35	74.92	79.81
C9	16.70	60.29	57.40	44.43	71.39	50.16	60.76	70.27	68.55	71.83	**73.19**
C10	15.11	32.83	23.90	34.76	44.54	30.51	31.52	39.77	42.58	46.01	**52.63**
C11	14.82	40.41	37.44	36.43	46.11	43.66	52.17	59.16	59.46	58.05	**61.53**
C12	31.30	55.22	45.75	53.44	60.13	24.52	48.03	59.96	59.55	69.44	**75.73**
C13	15.33	38.28	39.94	38.15	44.70	18.12	42.91	46.05	48.54	53.95	**60.99**
C14	76.20	79.43	81.23	89.73	84.96	82.79	83.34	**92.86**	92.24	90.82	91.86
C15	87.50	90.23	93.82	90.90	93.60	95.76	96.05	97.51	97.23	97.92	**98.42**
C16	73.04	82.01	85.48	72.51	85.34	84.03	86.99	86.73	89.63	91.13	**92.68**
C17	93.07	96.73	**100.00**	**100.00**	**100.00**	94.77	98.10	**100.00**	**100.00**	99.73	99.86
C18	48.42	77.83	79.21	77.03	85.39	88.48	86.59	**93.12**	92.14	92.11	92.69
C19	61.39	80.49	85.25	88.27	92.86	84.10	86.00	**95.54**	94.54	94.54	93.81
C20	81.16	90.54	94.25	89.52	96.22	96.07	95.82	97.77	97.02	98.15	**98.41**
OA	33.89	61.07	58.97	51.22	69.85	55.46	63.16	70.00	69.86	71.80	**74.37**
AA	63.55	77.27	77.44	72.67	82.55	76.29	80.18	84.25	84.60	85.45	**87.15**
κ	0.29	0.54	0.52	0.44	0.63	0.49	0.57	0.64	0.64	0.66	**0.69**

Gabor filtering utilizes both spatial information and spectral information of HSI, the classification accuracies obtained by 3DG-SVM are higher than 2DG-SVM, and 3DG-CNN performs better than 2DG-SVM as well.

More visually, Fig. 10 shows the training map, the testing map, and the classification maps of all these approaches in a single experiment on the Yellow River Estuary dataset. It can be seen from Fig. 10 that the classification maps generated by Gabor-related methods are smoother than the others. All RAW-SVM, CGDA-SVM, and LDA-SVM tend to have noisy classification maps. However, 2DG-SVM generates a messy result, and the classification map of 3DG-SVM seems to be oversmooth, which results in the loss of spatial structure properties. Alternatively, the classification map of our 3DG-CNN is both smooth and more consistent with the ground-truth map than all compared methods. All the observations validate the superiority of our methods and the rationality of integrating Gabor filtering and CNN module together.

In the following, detailed classification results on the Houston 2018 dataset are displayed in Table VI. Since the limitation of training samples (less than 1%), the classification accuracies of the Houston 2018 dataset are relatively lower than the Yellow River Estuary dataset, yet our proposed 3DG-CNN still exhibits the highest accuracies in most cases. Among the compared methods, RAW-SVM obtains extremely low classification accuracies without using any feature extraction strategy, and 3-D-Gabor-related methods perform better than 2-D-Gabor-related methods due to the spatial–spectral feature extraction. It should be noticed that 3DG-SVM outperforms all the other ML-based methods, together with 2DCNN and 3DCNN for a significant improvement and is competitive to MB-CNN, MS-CNN, and 2DG-CNN, which further validates the discriminative properties of the features spatially and spectrally extracted by 3-D Gabor filters. The training map, the testing map, and the predict maps obtained by various methods in a single experiment are visually provided

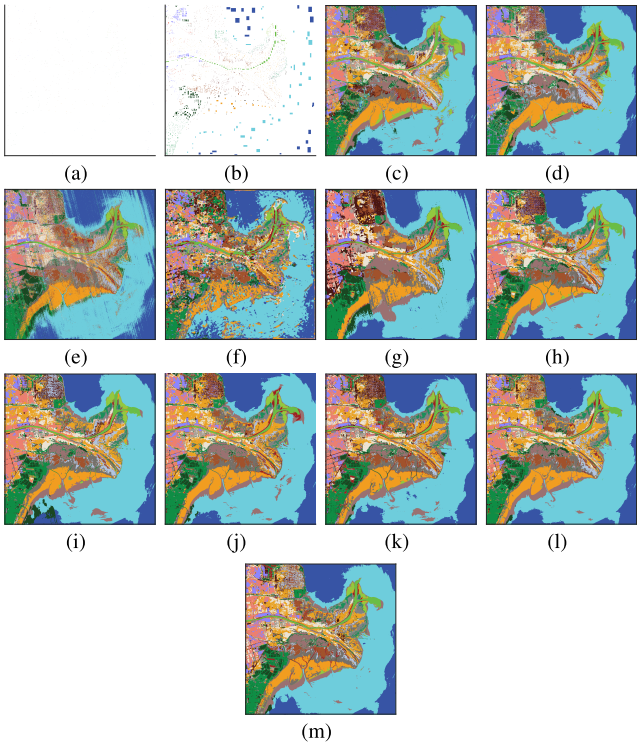

Fig. 10. (a) Training map, (b) testing map, and classification maps for the yellow river estuary dataset by (c) RAW-CNN (83.84%), (d) CGDA-SVM (87.50%), (e) LDA-SVM (85.96%), (f) 2DG-SVM (86.46%), (g) 3DG-SVM (89.80%), (h) 2DCNN (86.36%), (i) 3DCNN (89.11%), (j) MB-CNN (90.12%), (k) MS-CNN (92.22%), (l) 2DG-CNN (91.63%), and (m) 3DG-CNN (94.08%).

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TABLE VII
CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE FOR THE MATIWAN VILLAGE DATASET

Class	RAW-SVM	CGDA-SVM	LDA-SVM	2DG-SVM	3DG-SVM	2DCNN	3DCNN	MB-CNN	MS-CNN	2DG-CNN	3DG-CNN
C1	44.50	57.15	64.69	47.25	44.78	76.15	59.82	80.13	79.25	82.64	**86.55**
C2	51.43	68.23	70.55	64.97	62.28	84.97	74.93	85.14	86.50	87.58	**88.94**
C3	91.78	94.70	94.18	91.33	90.17	97.27	97.13	**98.99**	97.60	98.25	98.71
C4	90.90	95.36	94.71	72.30	91.23	96.27	96.96	97.13	97.28	97.41	**98.06**
C5	20.86	42.50	50.08	42.40	20.13	70.55	78.66	71.73	68.34	75.80	**79.59**
C6	56.99	59.12	71.32	52.69	70.41	75.04	75.97	79.87	81.13	85.11	**87.30**
C7	84.51	90.07	87.06	87.22	87.65	97.34	98.68	99.06	98.94	98.55	**99.34**
C8	92.45	86.08	91.93	85.58	89.39	93.45	94.78	93.55	95.03	95.55	**96.98**
C9	95.33	96.14	96.08	97.02	93.85	97.26	97.45	97.98	98.03	97.79	**98.94**
C10	92.22	93.17	94.95	74.15	92.33	96.48	95.67	97.58	97.44	97.47	**98.06**
C11	80.24	76.96	80.61	87.94	86.87	87.71	62.43	92.50	86.20	93.21	**95.69**
C12	56.95	68.68	69.64	72.66	50.18	84.98	80.74	87.47	87.46	88.66	**91.27**
C13	26.02	44.72	46.81	40.44	28.00	58.88	45.54	58.39	54.60	61.77	**67.25**
C14	74.51	78.99	80.79	93.13	77.41	90.78	87.41	93.46	92.12	94.14	**96.19**
C15	65.62	72.32	75.32	63.09	67.88	85.27	84.47	86.48	85.86	86.57	**88.81**
C16	43.74	45.20	58.89	75.72	47.54	68.83	63.26	71.28	67.20	78.08	**83.98**
C17	61.70	88.21	88.27	**99.94**	93.75	94.15	90.28	97.67	96.82	97.64	98.92
C18	52.29	62.06	67.46	44.43	59.82	71.08	70.26	76.23	77.88	76.70	**84.97**
C19	35.24	60.80	63.16	73.19	49.32	83.22	77.52	79.59	83.46	88.21	**91.55**
C20	74.99	83.52	88.57	95.76	80.46	87.83	87.51	92.69	90.53	90.69	**93.35**
OA	50.33	62.46	66.32	54.71	53.03	75.73	71.34	77.08	75.91	79.45	**83.26**
AA	64.61	73.20	76.75	73.06	69.17	84.88	80.97	86.85	86.08	88.59	**91.22**
κ	0.46	0.59	0.63	0.50	0.49	0.73	0.68	0.74	0.73	0.77	**0.82**

TABLE VIII
CLASSIFICATION ACCURACY (%) AND KAPPA MEASURE FOR THE PAVIA UNIVERSITY DATASET

Class	RAW-SVM	CGDA-SVM	LDA-SVM	2DG-SVM	3DG-SVM	2DCNN	3DCNN	MB-CNN	MS-CNN	2DG-CNN	3DG-CNN
C1	64.65	56.71	79.13	77.82	86.84	91.81	92.65	96.52	94.32	94.74	**96.52**
C2	64.31	74.77	81.26	83.76	87.97	91.44	92.44	87.82	88.62	92.32	**93.52**
C3	67.82	71.07	73.28	84.16	81.75	92.79	92.50	90.10	90.51	92.99	**94.49**
C4	93.98	92.71	95.08	87.27	91.10	96.89	95.35	**98.87**	97.93	96.74	96.92
C5	99.58	99.39	99.93	99.90	**100.00**	99.98	99.93	**100.00**	**100.00**	**100.00**	99.94
C6	70.44	75.60	86.51	80.78	**97.78**	90.63	85.14	89.76	90.52	92.42	92.39
C7	87.39	70.17	79.93	88.22	95.90	98.56	97.20	98.04	98.32	**99.05**	98.73
C8	75.23	64.65	72.61	78.06	70.66	93.69	89.99	91.80	93.97	95.56	**95.70**
C9	**99.88**	61.96	99.61	97.52	97.00	99.65	99.21	99.57	99.40	99.64	99.60
OA	70.62	72.60	82.26	83.12	88.15	92.65	92.10	91.50	91.75	93.88	**94.78**
AA	80.36	74.11	85.26	86.39	89.89	95.05	93.83	94.72	94.84	95.94	**96.42**
κ	0.63	0.63	0.77	0.78	0.85	0.91	0.90	0.89	0.90	0.92	**0.93**

in Fig. 11. As expected, 3DG-CNN generates a more realistic classification map, which is also smooth and in more accordance with the ground-truth map.

Next, concerning the Matiwan Village dataset, the quantitative and visualized results are, respectively, presented in Table VII and Fig. 12 in detail. Similarly, RAW-SVM obtains the lowest classification accuracies without feature extraction. Since the high interclass similarity of the Matiwan Village dataset, all DL-based methods with better abstract feature extraction capabilities outperform ML-based methods in the experiments. Besides, CGDA-SVM and LDA-SVM perform better results than 2DG-SVM and 3DG-SVM, which are more dependent on texture properties. In particular, with the integration of the texture properties representation capacity of 3-D Gabor filtering and the deep feature extraction ability of CNN, our proposed 3DG-CNN still achieves the best performance compared to all the other methods in OA, AA, Kappa, and accuracies in almost all land cover types (18 out of 20). Together with the smoothest and clearest classification map in Fig. 12, all the experiment results confirm the validity and reliability of the proposed method.

Finally, Table VIII reports the classification accuracies of all methods on the Pavia University dataset. It can be seen from Table VIII that RAW-SVM still provides the worst result. Besides, all the DL-based models perform better classification results than the ML-based methods, and 2DG-CNN and 3DG-CNN achieve higher accuracy than both 2DCNN and 3DCNN with scale and orientation information enhanced by Gabor filtering. Classification maps of the nine compared methods are presented in Fig. 13. It can be easily observed from Fig. 13 that our 3DG-CNN obtains not only the smoothest but also the most accurate map with the ground truth, which again proves the effectiveness of the proposed 3DG-CNN approach.

D. Experiments With Varying Number of Training Samples

Experiments with varying numbers of training samples on all datasets have been conducted to verify the robustness of

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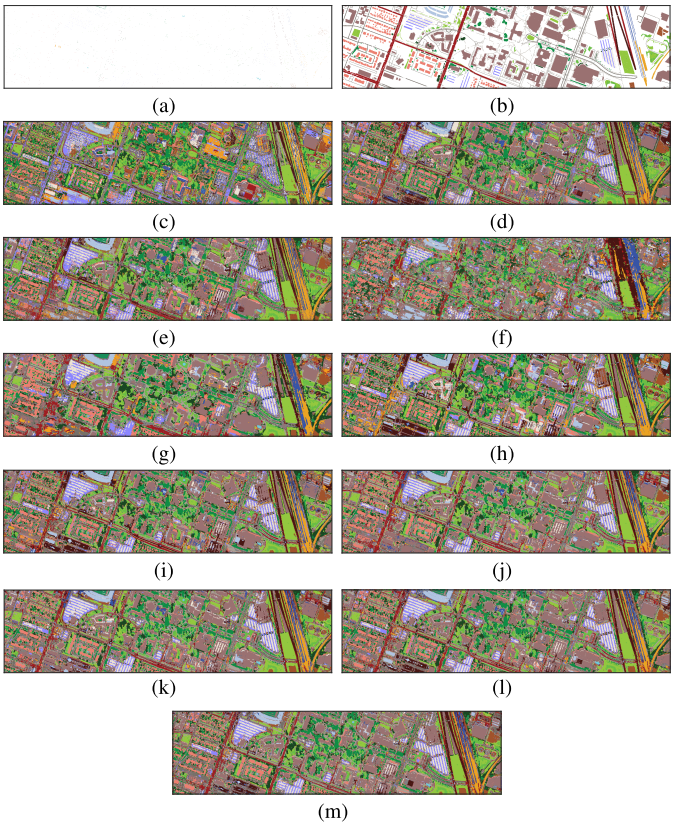

Fig. 11. (a) Training map, (b) testing map, and classification maps for the Houston 2018 dataset by (c) RAW-CNN (34.18%), (d) CGDA-SVM (62.72%), (e) LDA-SVM (60.09%), (f) 2DG-SVM (50.49%), (g) 3DG-SVM (68.76%), (h) 2DCNN (55.79%), (i) 3DCNN (62.84%), (j) MB-CNN (70.02%), (k) MS-CNN (72.57%), (l) 2DG-CNN (71.15%), and (m) 3DG-CNN (75.50%).

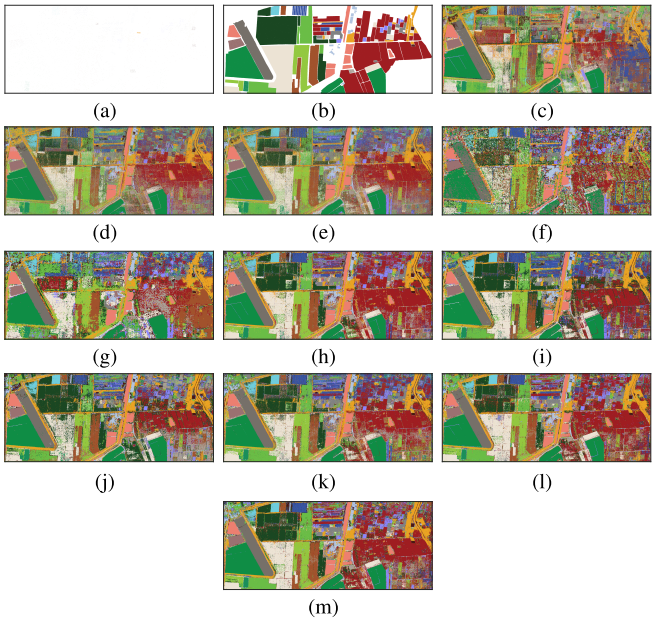

Fig. 12. (a) Training map, (b) testing map, and classification maps for the Matiwan Village dataset by (c) RAW-CNN (50.61%), (d) CGDA-SVM (63.16%), (e) LDA-SVM (67.03%), (f) 2DG-SVM (54.98%), (g) 3DG-SVM (54.00%), (h) 2DCNN (76.93%), (i) 3DCNN (72.62%), (j) MB-CNN (77.10%), (k) MS-CNN (77.30%), (l) 2DG-CNN (81.27%), and (m) 3DG-CNN (84.08%).

the proposed method. Precisely, the training samples of each dataset were randomly selected from all samples with six different percentages per class ten times, and the rest of the samples were used as the testing set. The experiment results were averaged by ten independent runs and are shown as OA, AA, and Kappa curves with standard deviation error bars in Figs. 14–17. It can be seen from the curves that our proposed 3DG-CNN always maintains the highest classification accuracy compared with the other methods with respect to various percentages of training samples. Experimental results show both the great classification performance and the robustness of our proposed 3DG-CNN to the varying of training samples.

E. Ablation Studies Toward Multibranch Structure

To verify the rationality of the multibranch structure ($U = 4$), it was compared with the single-/double-/triple-branch ($U = 1/2/3$) version of 3DG-CNN. For a fair comparison, all branches of the compared networks are consistent with the branches of 3DG-CNN, in which the two 3-D GMK convolutional layers of a single branch have the same scale of 3-D Gabor kernels. The scale factors used in the compared network with U branches were all the U-combinations of $f \in \{0.5, 0.25, 0.125, 0.0625\}$. Besides, all the networks were trained with the same training setting as mentioned in Section IV-C. As listed in each column of Table IX, the values of classification accuracy were averaged by U runs of U-branch network with different combinations of f.

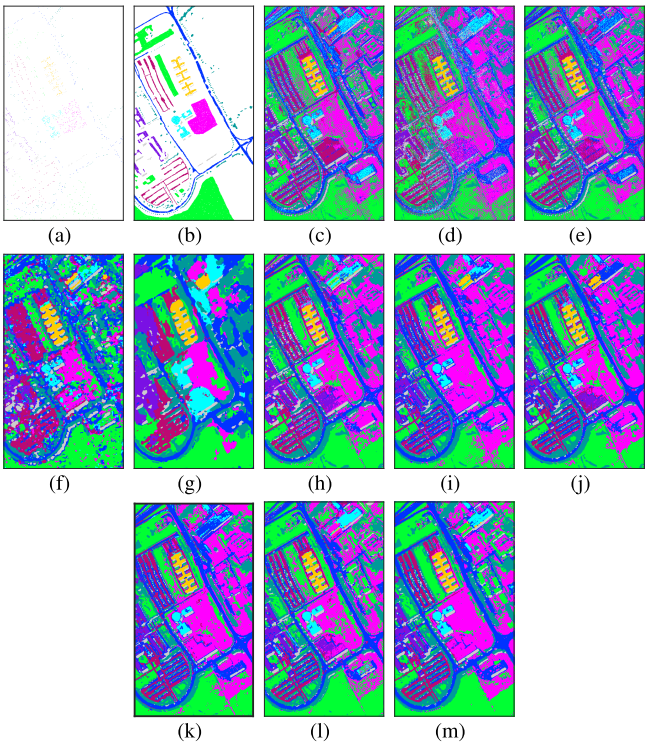

Fig. 13. (a) Training map, (b) testing map, and classification maps for the Pavia University dataset by (c) RAW-CNN (72.14%), (d) CGDA-SVM (72.60%), (e) LDA-SVM (84.01%), (f) 2DG-SVM (84.00%), (g) 3DG-SVM (89.64%), (h) 2DCNN (92.98%), (i) 3DCNN (92.35%), (j) MB-CNN (91.50%), (k) MS-CNN (92.19%), (l) 2DG-CNN (93.55%), and (m) 3DG-CNN (95.84%).

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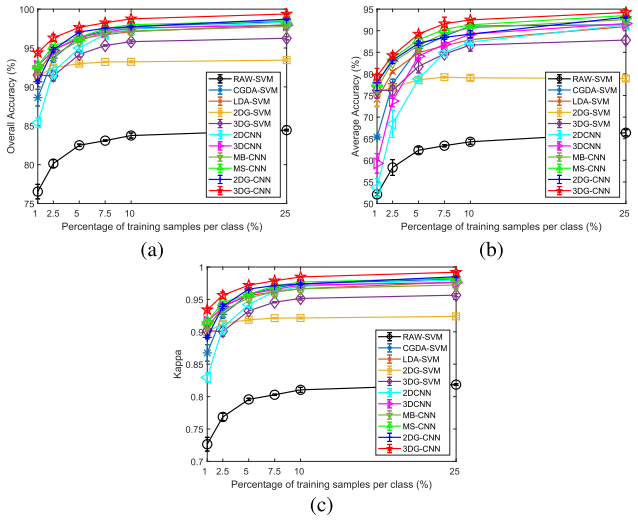

Fig. 14. Classification accuracy versus different percentage of training samples per class on the Yellow River Estuary dataset. (a) OA (%). (b) AA (%). (c) Kappa.

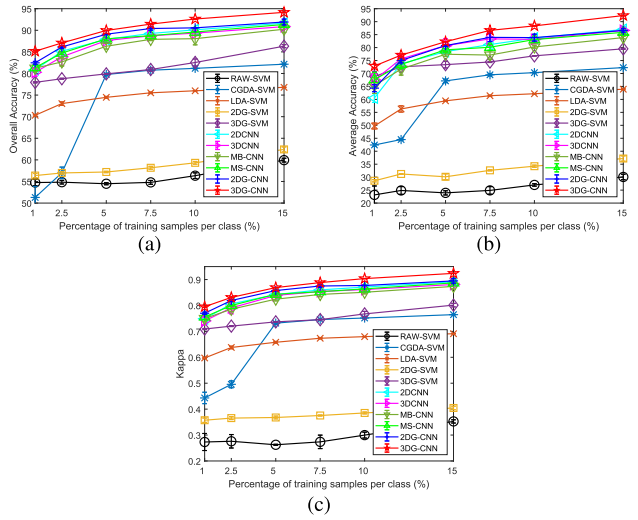

Fig. 15. Classification accuracy versus different percentage of training samples per class on the Houston 2018 dataset. (a) OA (%). (b) AA (%). (c) Kappa.

Experimental results consistently show that the multibranch structure of 3DG-CNN ($U = 4$) can extract more discriminative features by involving various scales from large to small.

F. Computation Cost Analysis

One of the motivations of 3DG-CNN is to relieve the complexity of neural network training by reducing the training parameters, so as to avoid the occurrence of overfitting. In our proposed 3DG-CNN approach, the capability of multiple 3-D Gabor kernels is integrated by learning a single 3-D SCK, which reduces the training parameters, while maintaining superior classification ability. Besides, the input of 3DG-CNN is normalized as independent patches of 13 bands, which reduces not only the parameters of the first layer to be trained but also the sliding operations of 3-D SCK in the spectral dimension, and further alleviates the complexity of training.

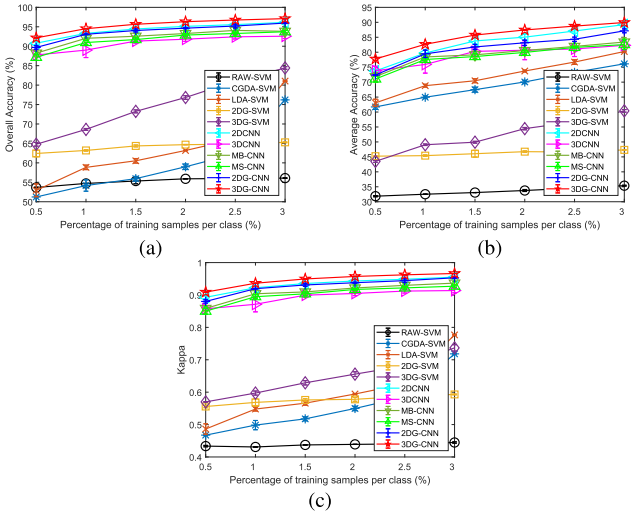

Fig. 16. Classification accuracy versus different percentage of training samples per class on the Matiwan Village dataset. (a) OA (%). (b) AA (%). (c) Kappa.

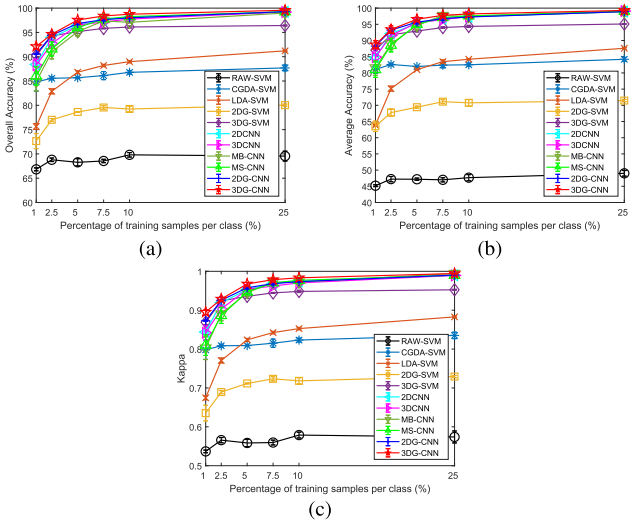

Fig. 17. Classification accuracy versus different percentage of training samples per class on the Pavia University dataset. (a) OA (%). (b) AA (%). (c) Kappa.

TABLE IX

ABLATION STUDY RESULTS TOWARD THE MULTIBRANCH STRUCTURE OF 3DG-CNN ON DIFFERENT DATASETS

Data Set	Accuracy	$U = 1$	$U = 2$	$U = 3$	$U = 4$
Yellow River Estuary	OA (%)	91.42	91.48	92.18	**92.75**
	AA (%)	87.56	88.56	89.20	**89.95**
	κ	0.90	0.90	0.92	**0.92**
Houston 2018	OA (%)	69.99	70.57	73.11	**74.37**
	AA (%)	85.14	86.35	87.00	**87.15**
	κ	0.65	0.67	0.68	**0.69**
Matiwan Village	OA (%)	80.50	81.84	82.60	**83.26**
	AA (%)	88.93	89.29	90.33	**91.22**
	κ	0.77	0.81	0.81	**0.82**
Pavia University	OA (%)	92.12	93.35	93.86	**94.78**
	AA (%)	93.56	94.76	95.10	**96.42**
	κ	0.89	0.91	0.92	**0.93**

Due to the great differences among the structures of the compared DL-based methods, to facilitate a fair comparison, we, respectively, calculated the computation cost of a single 2-D SCK layer, 3-D SCK layer, 2-D GMK layer, and 3-D

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TABLE X
COMPARISON OF THE COMPUTATION CONSUMPTION ON DIFFERENT DATASETS

Data Set	Layers	Parameter Size	Memory (MB)	Time (s)
Yellow River Estuary	2-D SCK	161,280	0.62	**0.04**
	3-D SCK	483,840	1.85	0.24
	2-D GMK	645,120	2.46	1.10
	3-D GMK	**157,248**	**0.60**	0.13
Houston 2018	2-D SCK	27,648	0.11	**0.02**
	3-D SCK	82,944	0.32	0.15
	2-D GMK	110,592	0.42	0.26
	3-D GMK	**22,464**	**0.09**	0.03
Matiwan Village	2-D SCK	147,456	0.56	**0.04**
	3-D SCK	442,368	1.69	0.22
	2-D GMK	589,824	2.25	1.00
	3-D GMK	**142,272**	**0.54**	0.12
Pavia University	2-D SCK	59,328	0.23	**0.03**
	3-D SCK	177,984	0.68	0.19
	2-D GMK	237,312	0.91	0.40
	3-D GMK	**52,416**	**0.20**	0.05

GMK layer with the same layer width (setting to be 64) on all datasets. Quantitatively, the parameter size, memory burden, and computation time of these single layers are reported in Table X. It is clearly shown that the 3-D GMK layer is more parameter saving than the others, which is of less training complexity. As for computation time, the 3-D GMK layer is second only to the 2-D SCK layer even though what were learned in the training procedure of the 3-D GMK layer are actually 3-D SCKs.

V. CONCLUSION

In this article, we present a 3DG-CNN for HSI classification. 3-D Gabor filters of multiscales and multiorientations are introduced to modulate the original SCKs of CNNs, so as to guide the CNN module to generate features with an enhanced robustness against the orientation and scale changes. The multibranch network architecture was later adopted to further enhance the performance of 3-D GMKs in multiple scales and orientations.

In the experiments, five ML-based methods and five DL-based methods are adopted as compared methods. We use four real-world HSI datasets to evaluate the performance of the methods, including the Yellow River Estuary, Houston 2018, Matiwan Village, and Pavia University datasets. Both experiments with fixed and varying numbers of training samples were conducted. The proposed 3DG-CNN always achieves the best performance on all datasets and is more parameter saving than the others, which demonstrates the superiority and effectiveness of 3DG-CNN. Ablation studies toward the multibranch structure further verify the rationality of our proposed method. Moreover, in order to facilitate the related research in the field, the source code of the proposed 3DG-CNN approach will be released in public.

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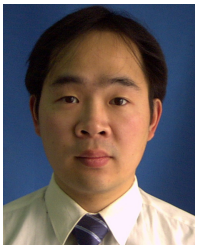

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